

Harmonic Function & Conjugate Harmonic Function

For an analytic function,

$f(z) = u + iv$, u or v are called harmonic functions if they satisfy Laplace's Equation,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow \nabla^2 u = 0$$

$$\text{or } \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \Rightarrow \nabla^2 v = 0$$

$$\text{where, } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

The real part u of an analytic function $f(z) = u + iv$ is known as the conjugate harmonic function of v and vice-versa.

Note: The real and imaginary parts of analytic functions are called conjugate harmonic functions.

First Method:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

If u is known, using the C-R eq^{ns}, (i.e. $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$)

$$f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

Integrating both sides w.r.to z ,

$$f(z) = \int \left(\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \right) dz.$$

Second Method:

Using the total differential formula for $v(x, y)$,

$$dv = \frac{\partial v}{\partial x} \cdot dx + \frac{\partial v}{\partial y} \cdot dy$$

By the C-R equation,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Substituting these back into the differential gives,

$$dv = -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$$

Integrating both sides,

$$v = \int \left(-\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \right) + C.$$

Theorem

The real and imaginary parts of an analytic function are harmonic.

Proof:

Let $f(z) = u + iv$ be an analytic function.

Since, $f(z)$ is analytic, it satisfies the C-R equation,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{--- (1)}$$

$$\& \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{--- (2)}$$

Differentiating eqⁿ. (1) w.r.to x and eqⁿ. (2) w.r.to y ,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{--- (3)}$$

$$\& \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x} = -\frac{\partial^2 v}{\partial x \partial y} \quad \text{--- (4)}$$

Adding eqⁿ. (3) & (4),

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial x \partial y}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$\therefore u$ is a harmonic function.

Similarly, on differentiating eqⁿ. (1) w.r.to y and eqⁿ. (2) w.r.to x , and adding them,

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = - \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial x \partial y}$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

$\therefore v$ is a harmonic function.

Q. Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic and find its harmonic conjugate function $v(x, y)$.

Solⁿ:

$$\text{Given, } u = \frac{1}{2} \log(x^2 + y^2) \text{ — (1)}$$

Diff. w.r.to x ,

$$\frac{\partial u}{\partial x} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2} \cdot 2x = \frac{x}{x^2 + y^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{(x^2 + y^2)(1) - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \text{ — (2)}$$

Again diff. eqⁿ. (1) w.r.to y ,

$$\frac{\partial u}{\partial y} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2} \cdot 2y = \frac{y}{x^2 + y^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{(x^2 + y^2)(1) - y(2y)}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \text{ — (3)}$$

Adding eqⁿ. (2) & (3), we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{y^2 - x^2 + x^2 - y^2}{(x^2 + y^2)^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Hence, u is a harmonic function.

For the harmonic conjugate $v(x, y)$,

By the C-R equations,

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \& \quad \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$

$$\Rightarrow dv = -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$$

$$\Rightarrow dv = -\left(\frac{y}{x^2+y^2}\right) dx + \left(\frac{x}{x^2+y^2}\right) dy$$

$$\Rightarrow dv = \frac{x dy - y dx}{x^2+y^2}$$

Here, the RHS is differential of $\tan^{-1}(y/x)$,

$$d\left(\tan^{-1}\left(\frac{y}{x}\right)\right) = \frac{1}{1+(y/x)^2} \cdot \frac{x dy - y dx}{x^2}$$

$$= \frac{x dy - y dx}{x^2+y^2}$$

Integrating dv ,

$$\int dv = \int d\left(\tan^{-1}\left(\frac{y}{x}\right)\right)$$

$$\Rightarrow v = \tan^{-1}\left(\frac{y}{x}\right) + C.$$

Q. Show that $v = \sinh x \cdot \cos y$ is harmonic and find its harmonic conjugate u .

Solⁿ:

$$\text{Given, } v = \sinh x \cdot \cos y \text{ — (1)}$$

$$\Rightarrow \frac{\partial v}{\partial x} = \cosh x \cdot \cos y$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} = \sinh x \cdot \cos y \text{ — (2)}$$

$$\text{Again, } \frac{\partial v}{\partial y} = -\sinh x \cdot \sin y$$

$$\Rightarrow \frac{\partial^2 v}{\partial y^2} = -\sinh x \cdot \cos y \text{ — (3)}$$

Adding (2) & (3),

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \sinh x \cdot \cos y - \sinh x \cdot \cos y$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

$\therefore v$ is harmonic.

Now,

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$\Rightarrow du = \frac{\partial v}{\partial y} dx - \frac{\partial v}{\partial x} dy \left[\because \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \right. \\ \left. \& \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \right]$$

$$\Rightarrow du = (-\sinh x \cdot \sin y) dx$$

$$- (\cosh x \cdot \cos y) dy$$

$$\Rightarrow du = - [(\sinh x \cdot \sin y) dx + (\cosh x \cdot \cos y) dy]$$

$$\Rightarrow \int du = - \int d(\cosh x \cdot \sin y)$$

$$\Rightarrow u = -\cosh x \cdot \sin y + C.$$

Milne's Thomson Method

This method is used to find any analytic function $f(z)$ when:

1. u is given.
2. v is given.
3. $u+v$ is given.
4. $u-v$ is given.

where $f(z) = u + iv$.

1. When u is given:

(i) $f(z) = u + iv$

(ii) $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \quad \left[\because \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \right]$$

(iii) Let, $\frac{\partial u}{\partial x} = \theta_1(x, y)$ & $\frac{\partial u}{\partial y} = \theta_2(x, y)$.

(iv) $f'(z) = \theta_1(x, y) - i \theta_2(x, y)$

(v) Replace x by z & y by 0 in $f'(z)$.

(vi) $f'(z) = \theta_1(z, 0) - i \theta_2(z, 0)$
 $\Rightarrow f(z) = \int [\theta_1(z, 0) - i \theta_2(z, 0)] dz + C$

2. When v is given:

(i) $f(z) = u + iv.$

(ii) $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$

$$\Rightarrow f'(z) = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x} \left[\because \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \right]$$

(iii) Let, $\frac{\partial v}{\partial y} = \psi_2(x, y)$ & $\frac{\partial v}{\partial x} = \psi_1(x, y).$

(iv) $f'(z) = \psi_2(x, y) + i \psi_1(x, y).$

(v) Replace x by z & y by 0 in $f'(z).$

(vi) $f'(z) = \psi_2(z, 0) + i \psi_1(z, 0)$

$$\Rightarrow f(z) = \int [\psi_2(z, 0) + i \psi_1(z, 0)] dz + C.$$

3. When $u+iv$ is given:

(i) $f(z) = u + iv$

$$\Rightarrow if(z) = iu - v$$

$$\Rightarrow (1+i)f(z) = (u-v) + i(u+v)$$

Let, $(1+i)f(z) = F(z)$, $u-v = U$ & $u+v = V$.

$$F(z) = U + iV.$$

(ii) $F'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x}$

$$\Rightarrow F'(z) = \frac{\partial V}{\partial y} + i \frac{\partial V}{\partial x} \left[\because \frac{\partial U}{\partial x} = \frac{\partial V}{\partial y} \right]$$

(iii) Let, $\frac{\partial V}{\partial y} = \psi_2(x, y)$ & $\frac{\partial V}{\partial x} = \psi_1(x, y)$.

(iv) $F'(z) = \psi_2(x, y) + i\psi_1(x, y)$

(v) Replace x by z & y by 0 .

(vi) $F'(z) = \psi_2(z, 0) + i\psi_1(z, 0)$

(vii) $F(z) = \int [\psi_2(z, 0) + i\psi_1(z, 0)] dz + C$

$$\Rightarrow f(z) = \frac{F(z)}{1+i}$$

4. When $u-v$ is given:

(i) $f(z) = u + iv$

$$\Rightarrow if(z) = iu - v$$

$$\Rightarrow (1+i)f(z) = (u-v) + i(u+v)$$

$$\Rightarrow F(z) = U + iV$$

Now, U is given.

(ii) $F'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x}$

$$\Rightarrow F'(z) = \frac{\partial U}{\partial x} - i \frac{\partial U}{\partial y} \left[\because \frac{\partial V}{\partial x} = -\frac{\partial U}{\partial y} \right]$$

(iii) Let, $\frac{\partial U}{\partial x} = \phi_1(x, y)$ & $\frac{\partial U}{\partial y} = \phi_2(x, y)$.

$$\Rightarrow F'(z) = \phi_1(x, y) - i \phi_2(x, y)$$

(iv) Replace x by z & y by 0 .

(v) $F'(z) = \phi_1(z, 0) - i \phi_2(z, 0)$.

(vi) $F(z) = \int [\phi_1(z, 0) - i \phi_2(z, 0)] dz + C$

$$\Rightarrow f(z) = \frac{F(z)}{1+i}$$

Q. If $u = e^x(x \cos y - y \sin y)$ is a harmonic function, find an analytic function $f(z) = u + iv$ such that $f(1) = e$.

Solⁿ:

$$f(z) = u + iv$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \left[\because \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \right]$$

Given, $u = e^x(x \cos y - y \sin y)$,

$$\frac{\partial u}{\partial x} = e^x \cos y + e^x(x \cos y - y \sin y)$$

$$\frac{\partial u}{\partial y} = e^x(-x \sin y - y \cos y - \sin y)$$

$$f'(z) = \left[e^x \cos y + e^x(x \cos y - y \sin y) - i[e^x(-x \sin y - y \cos y - \sin y)] \right]$$

Replace x by z & y by 0 ,

$$f'(z) = e^z + z e^z$$

Integrating both sides w.r.to z ,

$$f(z) = \int e^z(1+z) dz + C$$

$$\Rightarrow f(z) = \int e^z dz + \int z e^z dz + C$$

$$\Rightarrow f(z) = e^z + (z e^z - e^z) + C$$

$$\Rightarrow f(z) = z e^z + C$$

Also given, $f(1) = e$

$$\therefore f(1) = e$$

$$\therefore f(1) = 1 \cdot e^1 + C$$

$$\Rightarrow f(1) = e + C$$

$$\Rightarrow e = e + C$$

$$\Rightarrow C = 0$$

Hence, $f(z) = z e^z$.

9. Determine analytic function $f(z)$ whose real part is $e^{-x}(x \sin y - y \cos y)$.

Solⁿ:

Given, $u = e^{-x}(x \sin y - y \cos y)$

$$f(z) = u + iv$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$\left[\because \frac{\partial v}{\partial x} = - \frac{\partial u}{\partial y} \right]$$

Now,

$$\frac{\partial u}{\partial x} = e^{-x} \frac{\partial}{\partial x} (x \sin y - y \cos y) + (x \sin y - y \cos y) \cdot \frac{\partial}{\partial x} (e^{-x})$$

$$\Rightarrow \frac{\partial u}{\partial x} = e^{-x} (\sin y) + (x \sin y - y \cos y) \cdot (-e^{-x})$$

$$\Rightarrow \frac{\partial u}{\partial x} = e^{-x} \cdot \sin y - x e^{-x} \sin y + y e^{-x} \cos y$$

$$\frac{\partial u}{\partial y} = e^{-x} \frac{\partial}{\partial y} (x \sin y - y \cos y)$$

$$\Rightarrow \frac{\partial u}{\partial y} = e^{-x} (x \cos y + y \sin y - \cos y)$$

$$f'(z) = e^{-x}(\sin y) - e^{-x}(x \sin y - y \cos y) \\ - i[e^{-x}(x \cos y + y \sin y - \cos y)]$$

Replace x by z & y by 0 ,

$$f'(z) = e^{-z}(\sin 0) - e^{-z}(z \sin 0 - 0 \cos 0) \\ - i[e^{-z}(z \cos 0 + 0 \sin 0 - \cos 0)]$$

$$\Rightarrow f'(z) = -i(z e^{-z} - e^{-z})$$

$$\Rightarrow f(z) = -i \left[\int z e^{-z} dz - \int e^{-z} dz \right]$$

$$\Rightarrow f(z) = -i \left[-z e^{-z} - e^{-z} + e^{-z} \right] + C$$

$$\Rightarrow f(z) = i z e^{-z} + C.$$

Q. Find analytic function whose imaginary part is given as $v = e^{-x}(x \cos y + y \sin y)$.

Solⁿ:

Given, $v = e^{-x}(x \cos y + y \sin y)$

$$f(z) = u + iv$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$

Now,

$$\frac{\partial v}{\partial y} = e^{-x} \cdot \frac{\partial}{\partial y} (x \cos y + y \sin y)$$

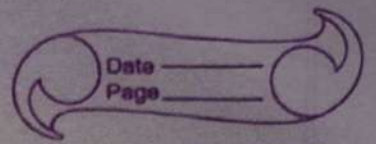
$$\Rightarrow \frac{\partial v}{\partial y} = e^{-x} (-x \sin y + \sin y + y \cos y)$$

$$\frac{\partial v}{\partial x} = e^{-x} \cdot \frac{\partial}{\partial x} (x \cos y + y \sin y) +$$

$$(x \cos y + y \sin y) \cdot \frac{\partial (e^{-x})}{\partial x}$$

$$\Rightarrow \frac{\partial v}{\partial x} = e^{-x} (\cos y) - e^{-x} (x \cos y + y \sin y)$$

$$\boxed{\begin{aligned}\int z e^z &= z e^z - e^z \\ \int z e^{-z} &= -z e^{-z} - e^{-z}\end{aligned}}$$



So, $f'(z) = e^{-x}(-x \sin y + \sin y + y \cos y)$
 $+ i[e^{-x}(\cos y) - e^{-x}(x \cos y + y \sin y)]$

~~$f'(z)$~~ $\rightarrow$

Replacing x by z & y by 0 ,

$$f'(z) = e^{-z}(-z \sin 0 + \sin 0 + 0 \cdot \cos 0)$$

$$+ i[e^{-z}(\cos 0) - e^{-z}(z \cos 0 + 0 \cdot \sin 0)]$$

$$\Rightarrow f'(z) = i(e^{-z} - z e^{-z})$$

$$\Rightarrow f(z) = i \left[\int e^{-z} dz - \int z e^{-z} dz \right] + C$$

$$\Rightarrow f(z) = i \left[-e^{-z} - (-z e^{-z} - e^{-z}) \right] + C$$

$$\Rightarrow f(z) = i(-\cancel{e^{-z}} + z e^{-z} + \cancel{e^{-z}}) + C$$

$$\Rightarrow f(z) = i z e^{-z} + C.$$

Q. Show that the function $u = x^3 - 3xy^2$ is harmonic and find the corresponding analytic function.

Solⁿ:

Given, $u = x^3 - 3xy^2$

Now,

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = 6x$$

$$\frac{\partial u}{\partial y} = -6xy$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = -6x$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0$$

Hence, u is harmonic.

We know,

$$f(z) = u + iv$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2$$

$$\frac{\partial u}{\partial y} = -6xy$$

$$\text{So, } f'(z) = (3x^2 - 3y^2) + i(6xy)$$

Replacing x by z & y by 0 ,

$$f'(z) = 3z^2 - 3 \times 0 + i(6 \times z \times 0)$$

$$\Rightarrow f'(z) = 3z^2$$

$$\Rightarrow f(z) = 3 \int z^2 dz + C$$

$$\Rightarrow f(z) = \frac{3z^3}{3} + C$$

$$\Rightarrow f(z) = z^3 + C.$$

Q. Show that the function $e^x \cdot \cos y$ is harmonic and find the analytic function of which it is real part.

Solⁿ:

$$u = e^x \cdot \cos y$$

Now,

$$\frac{\partial u}{\partial x} = e^x \cdot \cos y$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = e^x \cdot \cos y$$

$$\frac{\partial u}{\partial y} = -e^x \cdot \sin y$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = -e^x \cdot \cos y$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \cos y - e^x \cos y = 0$$

Hence, the function $e^x \cdot \cos y$ is harmonic.

We know,

$$f(z) = u + iv$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$\Rightarrow f'(z) = e^x \cdot \cos y + i e^x \cdot \sin y$$

Replacing x by z & y by 0 ,

$$f'(z) = e^z \cdot \cos 0 + i e^z \cdot \sin 0$$

$$\Rightarrow f'(z) = e^z$$

$$\Rightarrow f(z) = \int e^z dz + C$$

$$\Rightarrow f(z) = e^z + C.$$

Q. Find the regular function $f(z)$ whose imaginary part is $\cos x \cdot \cosh y$.

Solⁿ:

Given, $v = \cos x \cdot \cosh y$

We have,

$$f(z) = u + iv$$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$

Now,

$$\frac{\partial v}{\partial x} = -\sin x \cdot \cosh y$$

$$\frac{\partial v}{\partial y} = \cos x \cdot \sinh y$$

$$\text{So, } f'(z) = \cos x \cdot \sinh y - i (\sin x \cdot \cosh y)$$

Replacing x by z & y by 0 ,

$$f'(z) = \cos z \cdot \sinh 0 - i (\sin z \cdot \cosh 0)$$

$$\Rightarrow f'(z) = -i \sin z$$

$$\Rightarrow f(z) = -i \int \sin z \, dz + C$$

$$\Rightarrow f(z) = i \cos z + C.$$

Q. Find analytic function $f(z)$ if $u - v = e^{-x}[(x-y) \sin y - (x+y) \cos y]$.

Solⁿ:

$$\text{Given, } u - v = e^{-x}[(x-y) \sin y - (x+y) \cos y]$$

We have,

$$f(z) = u + iv$$

$$\Rightarrow if(z) = iu - v \quad \leftarrow [i^2 = -1] \quad [\text{Multiplying both sides by } i]$$

$$\Rightarrow (1+i)f(z) = (u-v) + i(u+v) \quad [\text{Adding both above eqns.}]$$

$$\text{Let, } (1+i)f(z) = F(z), \quad \Rightarrow f(z) = \frac{F(z)}{(1+i)}$$

$$u - v = U$$

$$\& \quad u + v = V$$

$$\text{So, } F(z) = U + iV$$

$$\Rightarrow F'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x}$$

Here, $U = u - v$ is given.

$$F'(z) = \frac{\partial U}{\partial x} - i \frac{\partial U}{\partial y}$$

Now,

$$\frac{\partial U}{\partial x} = [(x-y) \sin y - (x+y) \cos y] \cdot \frac{\partial}{\partial x} (e^{-x}) + e^{-x} \cdot \frac{\partial}{\partial x} [(x-y) \sin y - (x+y) \cos y]$$

$$\Rightarrow \frac{\partial U}{\partial x} = -e^{-x} [(x-y) \sin y - (x+y) \cos y] + e^{-x} [\sin y - \cos y]$$

And,

~~$$\frac{\partial U}{\partial y} = e^{-x} [x \cos y - (\sin y + y \cos y) + x \sin y - \cos y + y \sin y]$$~~

$$\frac{\partial U}{\partial y} = e^{-x} \left[(x-y) \cdot \frac{\partial}{\partial y} (\sin y) + \sin y \cdot \frac{\partial}{\partial x} (x-y) - (x+y) \cdot \frac{\partial}{\partial y} (\cos y) - \cos y \cdot \frac{\partial}{\partial x} (x+y) \right]$$

$$\Rightarrow \frac{\partial U}{\partial y} = e^{-x} [(x-y) \cos y - \sin y + (x+y) \sin y - \cos y]$$

$$\text{So, } F'(z) = e^{-x}(\sin y - \cos y) - e^{-x}[(x-y)\sin y - (x+y)\cos y] \\ - i e^{-x}[(x-y)\cos y - \sin y + (x+y)\sin y - \cos y]$$

Replace x by z & y by 0 ,

$$F'(z) = e^{-z}(\underbrace{\sin 0}_0 - \underbrace{\cos 0}_1) - e^{-z}[(z-0)\underbrace{\sin 0}_0 - (z+0)\underbrace{\cos 0}_1] \\ - i e^{-z}[(z-0)\underbrace{\cos 0}_1 - \underbrace{\sin 0}_0 + (z+0)\underbrace{\sin 0}_0 - \underbrace{\cos 0}_1]$$

$$\Rightarrow F'(z) = -e^{-z} + ze^{-z} - i[ze^{-z} - e^{-z}]$$

$$\Rightarrow F'(z) = -e^{-z} + ze^{-z} - i ze^{-z} + i e^{-z}$$

$$\Rightarrow F'(z) = -e^{-z}(1-i) + ze^{-z}(1-i)$$

$$\Rightarrow F'(z) = (1-i)[ze^{-z} - e^{-z}]$$

$$\Rightarrow F(z) = (1-i)\left[\int ze^{-z} dz - \int e^{-z} dz\right]$$

$$\Rightarrow F(z) = (1-i)[-ze^{-z} - e^{-z} + e^{-z}] + C$$

$$[\int ze^{-z} dz = -ze^{-z} - e^{-z}]$$

$$\Rightarrow F(z) = -(1-i)ze^{-z} + C$$

$$\Rightarrow F(z) = (i-1)ze^{-z} + C.$$

$$f(z) = \frac{F(z)}{(1+i)} = \frac{(i-1)}{(1+i)} z e^{-z} + \frac{C}{(1+i)}$$

$$\left[\begin{aligned} \frac{i-1}{1+i} &= \frac{(i-1)(i-1)}{(i+1)(i-1)} = \frac{(i-1)^2}{i^2-1^2} = \frac{1^2+i^2-2i}{-1-1} \\ &= \frac{1-1-2i}{-2} \\ &= -\frac{2i}{-2} \\ &= i \end{aligned} \right]$$

$$f(z) = i z e^{-z} + C'. \quad \left[C' = \frac{C}{(1+i)} \right]$$

Ans.

Q. Find analytic function $f(z)$ if $u+v = (x+y)(2-4xy+x^2+y^2)$.

Solⁿ:

$$\text{Given, } u+v = (x+y)(2-4xy+x^2+y^2)$$

We have,

$$f(z) = u + iv$$

$$\Rightarrow if(z) = iu - v$$

$$\Rightarrow (i+1)f(z) = (u-v) + i(u+v)$$

$$\text{Let, } F(z) = (i+1)f(z) \Rightarrow f(z) = \frac{F(z)}{i+1}$$

$$U = u - v \quad \& \quad V = u + v$$

$$F(z) = U + iV$$

Here, $V = u + v$ is given.

$$F'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x}$$

$$\Rightarrow F'(z) = \frac{\partial V}{\partial y} + i \frac{\partial V}{\partial x} \quad \left[\because \frac{\partial U}{\partial x} = \frac{\partial V}{\partial y} \right]$$

Now,

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} (x+y) [2-4xy+x^2+y^2] + (x+y) \cdot \frac{\partial}{\partial x} (2-4xy+x^2+y^2)$$

$$\Rightarrow \frac{\partial V}{\partial x} = (2-4xy+x^2+y^2) + (x+y)(-4y+2x)$$

And,

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y} (x+y) [2-4xy+x^2+y^2] + (x+y) \cdot \frac{\partial}{\partial y} (2-4xy+x^2+y^2)$$

$$\Rightarrow \frac{\partial V}{\partial y} = (2-4xy+x^2+y^2) + (x+y)(-4x+2y)$$

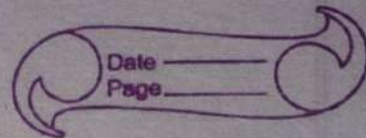
$$\text{So, } F'(z) = (x+y)(-4x+2y) + (2-4xy+x^2+y^2) + i [(x+y)(-4y+2x) + (2-4xy+x^2+y^2)]$$

Replace x by z & y by 0 ,

$$F'(z) = (z+0)(-4z+2 \times 0) + (2-4 \times z \times 0 + z^2 + 0^2) + i [(z+0)(-4 \times 0 + 2z) + (2-4 \times z \times 0 + z^2 + 0^2)]$$

$$\Rightarrow F'(z) = (-4z^2 + 2 + z^2) + i(2z^2 + 2 + z^2)$$

$$\left[\frac{i-1}{i+1} = i \right]$$



$$\Rightarrow F'(z) = (-3z^2 + 2) + i(3z^2 + 2)$$

$$\Rightarrow F(z) = \int [(-3z^2 + 2) + i(3z^2 + 2)] dz$$

$$\Rightarrow F(z) = -\frac{3z^3}{3} + 2z + i\left(\frac{3z^3}{3} + 2z\right) + C$$

$$\Rightarrow F(z) = -z^3 + 2z + iz^3 + 2iz + C$$

$$\Rightarrow F(z) = (i-1)z^3 + (1+i)2z + C$$

$$\Rightarrow f(z) = \frac{(i-1)}{(i+1)} z^3 + \frac{(1+i)}{(1+i)} 2z + \frac{C}{(1+i)} \quad \left[f(z) = \frac{F(z)}{(i+1)} \right]$$

$$\Rightarrow f(z) = iz^3 + 2z + \frac{C'}{(1+i)}$$

$$\Rightarrow f(z) = iz^3 + 2z + C' \quad \left[C' = \frac{C}{1+i} \right]$$

Q. Find analytic function $f(z)$ if $u-v = (x-y)(x^2+4xy+y^2)$.

Solⁿ:

Given, $u-v = (x-y)(x^2+4xy+y^2)$

$$f(z) = u + iv$$

$$\Rightarrow if(z) = iu - v$$

$$\begin{matrix} (1+i)f(z) & = & (u-v) + i(u+v) \\ F(z) & & U \quad \quad V \end{matrix}$$

$$F(z) = U + iV$$

$$\Rightarrow F'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x}$$

Here, $U = u-v$ is given.

$$F'(z) = \frac{\partial U}{\partial x} - i \frac{\partial U}{\partial y}$$

Now,

$$\frac{\partial U}{\partial x} = (x^2+4xy+y^2) \cdot \frac{\partial}{\partial x} (x-y) + (x-y) \cdot \frac{\partial}{\partial x} (x^2+4xy+y^2)$$

$$\Rightarrow \frac{\partial U}{\partial x} = (x^2+4xy+y^2) + (x-y)(2x+4y)$$

$$\frac{\partial U}{\partial y} = (x^2 + 4xy + y^2) \cdot \frac{\partial}{\partial y} (x-y) + (x-y) \frac{\partial}{\partial y} (x^2 + 4xy + y^2)$$

$$\Rightarrow \frac{\partial U}{\partial y} = (x^2 + 4xy + y^2) \times (-1) + (x-y)(4x + 2y)$$

So,

$$F'(z) = (x^2 + 4xy + y^2) + (x-y)(2x + 4y) - i[(x-y)(4x + 2y) - (x^2 + 4xy + y^2)]$$

Replace x by z & y by 0 ,

$$F'(z) = (z^2 + 4 \times z \times 0 + 0^2) + (z-0)(2 \times z + 4 \times 0)$$

$$- i[(z-0)(4 \times z + 2 \times 0) - (z^2 + 4 \times z \times 0 + 0^2)]$$

$$\Rightarrow F'(z) = (z^2 + 2z^2) - i(4z^2 - z^2)$$

$$\Rightarrow F'(z) = 3z^2 - 3iz^2$$

$$\Rightarrow F'(z) = 3z^2(1-i)$$

$$\Rightarrow F(z) = (1-i) \int 3z^2 dz$$

$$\Rightarrow F(z) = (1-i) \times \frac{3z^3}{3} + C$$

$$\Rightarrow F(z) = (1-i)z^3 + C$$

$$f(z) = \frac{F(z)}{(1+i)}$$

$$\Rightarrow f(z) = \frac{(1-i)}{(1+i)} z^3 + \frac{C}{(1+i)}$$

$$\left[\frac{1-i}{1+i} = \frac{(1-i)^2}{(1+i)(1-i)} = \frac{1^2 + i^2 - 2i}{1^2 - i^2} = \frac{1-1-2i}{1+1} = \frac{-2i}{2} = -i \right]$$

$$f(z) = -iz^3 + \frac{C}{(1+i)}$$

$$\Rightarrow f(z) = -iz^3 + C' \quad \left[C' = \frac{C}{1+i} \right]$$

Q. Find $f(z)$ if $u+v = \frac{2 \sin 2x}{e^{2y} + e^{-2y} + 2 \cos 2x}$.

Solⁿ:

Given, $u+v = \frac{2 \sin 2x}{e^{2y} + e^{-2y} + 2 \cos 2x}$

We know,

$$f(z) = u + iv$$

$$\Rightarrow if(z) = iu - v$$

$$\begin{matrix} (1+i)f(z) = (u-v) + i(u+v) \\ F(z) \qquad \qquad U \qquad \qquad V \end{matrix}$$

$$F(z) = U + iV$$

$$\Rightarrow F'(z) = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x}$$

Here, $V = uv$ is given.

$$F'(z) = \frac{\partial V}{\partial y} + i \frac{\partial V}{\partial x}$$

$$\left[\cosh y = \frac{e^y + e^{-y}}{2} \Rightarrow \cosh 2y = \frac{e^{2y} + e^{-2y}}{2} \right]$$

$$V = uv = \frac{2 \sin 2x}{e^{2y} + e^{-2y} + 2 \cos 2x}$$

$$\Rightarrow V = \frac{2 \sin 2x}{2 \cosh 2y + 2 \cos 2x}$$

$$\Rightarrow V = \frac{\sin 2x}{\cosh 2y + \cos 2x}$$

Now,

$$\frac{\partial V}{\partial x} = \frac{(\cosh 2y + \cos 2x) \cdot \frac{\partial}{\partial x}(\sin 2x) - \sin 2x \cdot \frac{\partial}{\partial x}(\cosh 2y + \cos 2x)}{(\cosh 2y + \cos 2x)^2}$$

$$\Rightarrow \frac{\partial V}{\partial x} = \frac{2(\cosh 2y + \cos 2x) \cdot \cos 2x + \sin^2(2x)}{(\cosh 2y + \cos 2x)^2}$$

$$\Rightarrow \frac{\partial V}{\partial x} = \frac{2 \cosh 2y \cdot \cos 2x + 2 \cos^2(2x) + 2 \sin^2(2x)}{(\cosh 2y + \cos 2x)^2}$$

$$\Rightarrow \frac{\partial V}{\partial x} = \frac{2 \cosh 2y \cdot \cos 2x + 2}{(\cosh 2y + \cos 2x)^2} \left[\sin^2 x + \cos^2 x = 1 \right]$$

And,

$$\frac{\partial V}{\partial y} = \frac{(\cosh 2y + \cos 2x) \cdot \frac{\partial}{\partial y}(\sin 2x) - \sin 2x \cdot \frac{\partial}{\partial y}(\cosh 2y + \cos 2x)}{(\cosh 2y + \cos 2x)^2}$$

$$\Rightarrow \frac{\partial V}{\partial y} = \frac{-2 \sin 2x \cdot \sinh 2y}{(\cosh 2y + \cos 2x)^2}$$

$$\text{So, } F'(z) = \frac{-2 \sin 2x \cdot \sinh 2y}{(\cosh 2y + \cos 2x)^2} + i \left[\frac{2 \cosh 2y \cdot \cos 2x + 2}{(\cosh 2y + \cos 2x)^2} \right]$$

Replace, x by z & y by 0 ,

$$F'(z) = \frac{-2 \sin 2z \cdot \sinh(2 \times 0)}{(\cosh 2 \times 0 + \cos 2z)^2} + i \left[\frac{2 \cosh(2 \times 0) \cdot \cos 2z + 2}{[\cosh(2 \times 0) + \cos 2z]^2} \right]$$

$$\Rightarrow F'(z) = 0 + i \left[\frac{2 \cos 2z + 2}{(1 + \cos 2z)^2} \right]$$

$$\Rightarrow F'(z) = 2i \left[\frac{\cancel{(1 + \cos 2z)}}{(1 + \cos 2z)^2} \right]$$

$$\Rightarrow F'(z) = \frac{2i}{1 + \cos 2z}$$

$$\Rightarrow F(z) = 2i \int \frac{dz}{1 + \cos 2z}$$

$$\Rightarrow F(z) = 2i \int \frac{dz}{2 \cos^2 z} \quad \left[(1 + \cos 2x) = 2 \cos^2 x \right]$$

$$\Rightarrow F(z) = i \int \sec^2 z \, dz$$

$$\Rightarrow F(z) = i \cdot \tan z + C$$

Therefore, $f(z) = \frac{F(z)}{1+i}$

$$\Rightarrow f(z) = \frac{i}{1+i} \cdot \tan z + \frac{C}{1+i}$$

$$\left[\frac{i}{1+i} = \frac{i(1-i)}{(1+i)(1-i)} = \frac{i-i^2}{1^2-i^2} = \frac{i+1}{1+1} = \frac{i+1}{2} \right]$$

$$f(z) = \frac{i+1}{2} \cdot \tan z + \frac{C}{1+i}$$

$$\Rightarrow f(z) = \left(\frac{i+1}{2} \right) \cdot \tan z + C' \quad \left[C' = \frac{C}{1+i} \right]$$